Universidad de los Andes

ISIS-2111 PLE
Project 3
Fecha: October 13, 2026

Task 1. For the third part of the project, your task is to reuse the Objectilang language definition implemented in Rascal for the second part of the project and build a full language with its types and type system. Specifically, starting from your syntax and AST for Objectilang you will have to:

1. Define the Parser for the language
2. Generate the code for the Objectilang definition from the syntax rules and the abstract syntax, so that you can run Objectilang programs. The input for the program generation should be a text file (`.ol`) with the contents of the program. You should show the result of the program in the console.
3. Install TypePal into the language
4. Update the grammar to include type annotations for program values (Integers, Booleans, Chars, Strings). Moreover object definitions and data structures defined should also include a (generic) type (given by the user)
5. Check that the types used in the program correspond to the definitions
6. Define a new rule to ensure the elements used in an object definition all exist, and verify that programs satisfy this new rule.

In the definition of the parser and the inclusion of the types into the language you may fix problems encountered in the previous iteration of the project. To fix the problems from the first part of the project you may follow the solutions provided by the TA's or start from the project's solution from other students. Together with the Rascal project of your working solution, you should hand in a small document explaining the changes done to the project and the source of the solution if you started from a previously working solution, explaining all changes with respect to your solution.

If you do not fix the problems from the previous iteration of the project (1) your language will continue not working, (2) you will be deducted points again for the same mistakes.

Hand in a `.zip` file with the solution and document.